The University of Texas at Austin
Practice Materials for In-Term Exam 2

M339G Predictive Analytics

November 13, 2026

Instructions: This is a closed book and closed notes exam. The maximal score on the exam is 100 points.
Time: 50 minutes

CONCEPTUAL QUESTIONS

Problem 2.1. (10 points) What is the crucial difference in the assumptions between LDA and QDA? What consequence does that change have on the classification criterion?

Solution. Solutions will vary. The salient point of any response which is to earn credit must be that while LDA assumes the same standard deviation for each subpopulation, the QDA does not. Thus, a quadratic term shows up in the discriminant for the QDA case as opposed to the linear expressions in the predictors in the LDA case.

Problem 2.2. (10 points) Explain how naive Bayes is different from LDA. Provide one advantage which stems from this different approach. Are there disadvantages?

Solution. Solutions will vary. The salient point of any response which is to earn credit must be that naive Bayes **assumes** independence between predictors and f does not in general assume a Gaussian distribution. In particular, naive Bayes can handle categorical predictors.

Problem 2.3. (10 points) Describe the differences and the similarities between LDA and the logistic regression for a two-class classification problem.

Solution. Solutions will vary. The salient point of any response which is to earn credit must be similar to the last couple of slides here Chapter 4 - LDA.

Problem 2.4. (10 points) Justify this statement from the textbook:

"Neither of QDA nor naive Bayes is a special case the other."

Solution. Solutions will vary. The salient point of any response which is to earn credit must be something along the lines of the second bullet point on page 160 from the textbook.

FREE RESPONSE PROBLEMS

Problem 2.5. (10 points) *Source: Pitman's "Probability."*
Heights and weights of a large group of people follow a bivariate normal distribution, with correlation 0.75. Of the people **at** the 90th percentile of weights, about what percentage are **above** the 90th percentile of heights?

Solution. Let (U, V) be the random pair which stands for the people's weights and heights in real units. Let (X, Y) be the random pair which stands for the people's weights and heights in standard units. Conditioning on U is f at the 90th percentile is (close to) equivalent to conditioning on $\{X = 1.28\}$. So, the probability that we are looking for is

$$\mathbb{P}[Y > 1.28 \mid X = 1.28].$$

Recall that $Y \mid X = x \sim N(\rho x, 1 - \rho^2)$. So, the probability we are seeking equals (with $x = 1.28$)

$$\mathbb{P}\left[Z > \frac{x - \rho x}{\sqrt{1 - \rho^2}}\right] = \mathbb{P}\left[Z > x\sqrt{\frac{1-\rho}{1+\rho}}\right]$$

With the given correlation coefficient of 0.75, our answer is

$$1 - \Phi\left(1.28\sqrt{\frac{1-0.75}{1+0.75}}\right) = 0.3142659$$

Problem 2.6. (15 points) *Source: An old SRM manual.*
For a regression tree, two nodes in the tree - defining regions R_1 and R_2 - have the following values of the response variable (on the training set):

$$R_1 : 2, 3, 3, 4, 5$$
$$R_2 : 2, 4, 6$$

The tree is pruned using cost complexity pruning. The split creating R_1 and R_2 is the optimal one to prune. Determine the smallest value of α for which this pruning will occur.

Solution. The mean of the values in R_1 is 3.4 and the average of values in R_2 is 4. Without the pruning, the total contribution to the RSS from these two nodes is

$$(2 - 3.4)^2 + 2(3 - 3.4)^2 + (4 - 3.4)^2 + (5 - 3.4)^2 + (2 - 4)^2 + (6 - 4)^2 = 13.2.$$

If the pruning happens, all the response values will end up in the same terminal node whose mean will be 3.625. That node's contribution to the RSS is

$$2(2 - 3.625)^2 + 2(3 - 3.625)^2 + 2(4 - 3.625)^2 + (5 - 3.625)^2 + (6 - 3.625)^2 = 13.875.$$

For any $\alpha \geq 13.875 - 13.2 = 0.675$, the pruning will occur.

Problem 2.7. (20 points) A classification tree is being constructed to predict whether an insured will file a claim. A random sample of 200 insureds contains 60 who filed a claim. You are considering two splits:

- Split A: One node has 50 observations with 30 claims and one node has 150 observations with 30 claims.
- Split B: One node has 70 observations with 40 claims and one node has 130 observations with 20 claims.

The total Gini index after a split is the weighted average of the Gini index at each node, with the weights proportional to the number of observations in each node. The total entropy after a split is the weighted average of the entropy at each node, with the weights proportional to the number of observations in each node.

Determine which of the following statements is/are true.

I. Split A is preferred based on the total Gini index.
II. Split B is preferred based on the total entropy.
III. Splits A and B result in the same number of classification errors.

a. I only
b. II only
c. I and III only

d. II and III only
e. The correct answer is not given by a., b., c., or d.

Solution. The correct answer is **d.**
Smaller values of the total Gini index and the total entropy are preferred. The claim proportions in the nodes are 0.6 and 0.2 for Split A, and $\frac{4}{7} \approx 0.5714$ and $\frac{2}{13} \approx 0.1538$ for Split B.

- I is false. Using $G = 2\hat{p}(1-\hat{p})$ at each node:
  - Split A: $0.25 \cdot 2(0.6)(0.4) + 0.75 \cdot 2(0.2)(0.8) = 0.12 + 0.24 = 0.36$.
  - Split B: $0.35 \cdot 2\left(\frac{4}{7}\right)\left(\frac{3}{7}\right) + 0.65 \cdot 2\left(\frac{2}{13}\right)\left(\frac{11}{13}\right) \approx 0.1714 + 0.1692 = 0.3407$.

  Split B has the smaller total Gini index.
- II is true. Using $D = -\hat{p}\ln\hat{p} - (1-\hat{p})\ln(1-\hat{p})$ at each node:
  - Split A: $0.25(0.6730) + 0.75(0.5004) \approx 0.1683 + 0.3753 = 0.5436$.
  - Split B: $0.35(0.6829) + 0.65(0.4293) \approx 0.2390 + 0.2791 = 0.5181$.

  Split B has the smaller total entropy. (The comparison is the same for any base of the logarithm.)
- III is true. Each node predicts its majority class.
  - Split A: 20 errors in the first node and 30 in the second, for 50 errors.
  - Split B: 30 errors in the first node and 20 in the second, for 50 errors.

  Both splits give 50 classification errors. Only the Gini index and the entropy can tell the two splits apart.

Problem 2.8. (20 points) Consider the following observations of (X, Y) with X being the predictor and Y being the response:

$$(1,4), \quad (2,6), \quad (3,5), \quad (6,1).$$

After one iteration of recursive binary splitting, there are two groups of observations. Find the members of the two groups.

Solution. Remember that - in general - the criterion for choosing the splits is to minimize the residual sum of squares (RSS)

$$\sum_{j=1}^{J} \sum_{i \in R_j} \left(y_i - \hat{y}_{R_j}\right)^2$$

where $\hat{y}_{R_j}$ stands for the average of the response variable in region R_j for $j = 1, ..., J$. However, since this problem is computationally too complex, we resort to **recursive** binary splitting. Hence, as there is one predictor only in our current problem, we must make the split along its possible values. Every available split is f binary and partitions the support of X into R and R^c. So, it creates an RSS with this structure

$$\sum_{i \in R} (y_i - \hat{y}_R)^2 + \sum_{i \in R^c} (y_i - \hat{y}_{R^c})^2.$$

In this problem, we can now proceed "by hand" from the lowest to the highest observed value of the predictor.

If $(1, 4)$ is the sole element in R, the mean response for the remaining points is

$$\frac{6+5+1}{3} = \frac{12}{3} = 4.$$

The RSS is

$$(6-4)^2 + (5-4)^2 + (1-4)^2 = 14\,.$$

If $(1,4)$ and $(2,6)$ form the first region, the mean response in that region is $\frac{10}{2} = 5$. The remaining points $(3,5)$ and $(6,1)$ are in the other region and their mean response is $\frac{6}{2} = 3$. So, the RSS is

$$(4-5)^2 + (6-5)^2 + (5-3)^2 + (1-3)^2 = 10\,.$$

If $(1,4)$, $(2,6)$, and $(3,5)$ are in the first region and only $(6,1)$ remains in the other region, then the average of the first region's values of the response variable is

$$\frac{4+6+5}{3} = 5.$$

So, the RSS equals

$$(4-5)^2 + (6-5)^2 + (5-5)^2 = 2.$$

Overall, the smallest RSS corresponds to the third partition with $(6,1)$ in its own region, and the remaining points in the other region.

Problem 2.9. (20 points) Consider the following observations of (X, Y) with X being the predictor and Y being the response:

$$(0,8), \quad (2,6), \quad (4,5), \quad (6,7).$$

After one iteration of recursive binary splitting, there are two groups of observations. Find the members of the two groups.

Solution. Remember that - in general - the criterion for choosing the splits is to minimize the residual sum of squares (RSS)

$$\sum_{j=1}^{J} \sum_{i \in R_j} \left(y_i - \hat{y}_{R_j}\right)^2$$

where $\hat{y}_{R_j}$ stands for the average of the response variable in region R_j for $j = 1, ..., J$. However, since this problem is computationally too complex, we resort to **recursive** binary splitting. Hence, as there is one predictor only in our current problem, we must make the split along its possible values. Every available split is f binary and partitions the support of X into R and R^c. So, it creates an RSS with this structure

$$\sum_{i \in R} (y_i - \hat{y}_R)^2 + \sum_{i \in R^c} (y_i - \hat{y}_{R^c})^2.$$

In this problem, we can now proceed "by hand" from the lowest to the highest observed value of the predictor.

If $(0,8)$ is the sole element in R, the mean response for the remaining points is

$$\frac{6+5+7}{3} = \frac{18}{3} = 6.$$

The RSS is

$$(6-6)^2 + (5-6)^2 + (7-6)^2 = 2\,.$$

If $(0,8)$ and $(2,6)$ form the first region, the mean response in that region is $\frac{14}{2} = 7$. The remaining points $(4,5)$ and $(6,7)$ are in the other region and their mean response is $\frac{12}{2} = 6$. So, the RSS is

$$(8-7)^2 + (6-7)^2 + (5-6)^2 + (7-6)^2 = 4\,.$$

If $(0,8)$, $(2,6)$, and $(4,5)$ are in the first region and only $(6,7)$ remains in the other region, then the average of the first region's values of the response variable is

$$\frac{8+6+5}{3} = \frac{19}{3}.$$

So, the RSS equals

$$\left(8-\frac{19}{3}\right)^2 + \left(6-\frac{19}{3}\right)^2 + \left(5-\frac{19}{3}\right)^2 = \frac{14}{3}.$$

Overall, the smallest RSS corresponds to the first partition with $(0, 8)$ in its own region, and the remaining points in the other region.

MULTIPLE CHOICE QUESTIONS

Problem 2.10. (5 points) *Source: SRM Sample Problem #29.*
Determine which of the following considerations may make decision trees preferable to other statistical methods.

I. Decision trees are easily interpretable.
II. Decision trees can be displayed graphically.
III. Decision trees are easier to explain than linear regression models.

a. None.
b. I and II only.
c. I and III only.
d. II and III only.
e. The correct answer is not given above.

Solution. (e)
All three statements are true.

Problem 2.11. (5 points) Determine which of the following statements about K-means clustering are true.

I. Each iteration of the K-means algorithm can never increase the total within-cluster variation.
II. The K-means algorithm is guaranteed to find the partition of the observations which globally minimizes the total within-cluster variation.
III. The clusters obtained can change if the variables are standardized before running the algorithm.

a. None.
b. I and II only.
c. I and III only.
d. II and III only.
e. The correct answer is not given above.

Solution. (c)
I is **true** as both reassigning each observation to the nearest centroid and recomputing the centroids as cluster means can only decrease or keep the value of the within-cluster sum of squares.
II is false as the algorithm only finds a local optimum, so the result depends on the initial random assignment. This is why it should be run several times from different starting points (recall `nstart` from `R`).
III is **true** since K-means uses Euclidean distance, so the clustering depends on the scale of the variables.

Problem 2.12. (5 points) Determine which of the following statements about choosing the number of clusters K in K-means clustering are true.

I. Choosing K to minimize the total within-cluster variation is not a sensible criterion.
II. One sensible heuristic for choosing K would be to plot the total within-cluster variation against K and pick the value of K after which further increases give only small reductions.
III. The clusters obtained with $K = 3$ can always be obtained by splitting one of the clusters obtained with $K = 2$.

a. None.
b. I and II only.
c. I and III only.
d. II and III only.
e. The correct answer is not given above.

Solution. (b)
I is **true** because as K grows, the smallest achievable total within-cluster variation never goes up, and it reaches 0 at $K = n$, when every observation is its own cluster. Minimizing it would therefore always pick the largest possible K.
II is **true** this is the "elbow" method. It looks for the point where adding clusters stops meaningfully reducing the within-cluster variation. Also, recall our reasoning when we considered scree plots.
III is false: K-means clusters are not nested in general. For example, the best two-cluster split of a population might separate it by gender, while the best three-cluster split might separate it by nationality.

Problem 2.13. (5 points) Determine which of the following statements about classification trees are true.

I. When growing a classification tree, the classification error rate is preferred over the Gini index as the splitting criterion because it is more sensitive to node purity.
II. A split may produce two terminal nodes which have the same predicted class.
III. The Gini index of a node is small when most of the observations in that node belong to a single class.

a. None.
b. I and II only.
c. I and III only.
d. II and III only.
e. The correct answer is not given above.

Solution. (d)
I is false. The truth is the other way around. The Gini index and entropy are more sensitive to node purity than the classification error rate, so they are preferred for growing the tree. The classification error rate is preferable for pruning when prediction accuracy is the goal.
II is **true**. Such a split does not change the predictions, but it can increase node purity, so it can still lower the Gini index or entropy.
III is **true** as the Gini index is $\sum_{k=1}^{K} \hat{p}_{mk}(1 - \hat{p}_{mk})$, by definition. Each term is close to 0 when $\hat{p}_{mk}$ is close to 0 or 1, so the index is small when a node is nearly pure.

The above problems are **in addition** to your past homework assignments and in-class work. Do not forget to re-solve those!